DATA ACQUISITION, CLEANING, AND EXPLORATORY ANALYSIS

Internship Task – Week 1

Project Title: Milk Quality Analysis Using Python

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Domain: Data Science

Tools Used: Python, Google Colab, Pandas, NumPy, Matplotlib, Seaborn

1. Introduction

Data preparation is one of the most important stages of a Data Science project. Before applying statistical analysis or machine learning algorithms, raw data must be collected, examined, cleaned, and explored.

For this task, a Milk Quality Dataset was selected for analysis. The dataset contains different physical and sensory characteristics of milk that can be used to understand and classify its quality.

Python was used to perform data acquisition, data inspection, cleaning, preprocessing, and exploratory data analysis. The analysis was performed using libraries such as Pandas, NumPy, Matplotlib, and Seaborn.

The main purpose of this task was to understand the dataset, identify data-quality issues, prepare the data for further analysis, and discover initial patterns through exploratory analysis.

2. Objectives

The main objectives of this task were:

1. To acquire a publicly available dataset relevant to a real-world problem.
2. To understand the structure and characteristics of the dataset.
3. To inspect data types and identify data-quality issues.
4. To check and handle missing values.

5. To identify and handle duplicate records.
6. To ensure that the variables are stored in appropriate data types.
7. To perform exploratory data analysis using summary statistics.
8. To create meaningful visualizations.
9. To identify important patterns and relationships in the data.
10. To prepare the dataset for further statistical analysis and machine learning.

3. Data Acquisition

A publicly available Milk Quality Dataset was selected for this project. The dataset was loaded into Python using the Pandas library.

The dataset contains milk quality characteristics such as pH, temperature, taste, odor, fat, turbidity, colour, and grade.

3.1 Dataset Loading

```
import pandas as pd
import numpy as np
df = pd.read_csv("milknew.csv")
df.head()
```

The "read_csv()" function was used to load the dataset into a Pandas DataFrame. The "head()" function was then used to display the first five records and understand the structure of the data.

4. Dataset Description

The dataset contains the following variables:

1. pH
2. Temprature
3. Taste

4. Odor

5. Fat

6. Turbidity

7. Colour

8. Grade

The first seven variables represent milk quality characteristics, while Grade represents the overall quality classification of the milk.

5. Data Inspection

After loading the dataset, its structure was examined using Pandas functions.

5.1 Dataset Shape

```
df.shape
```

The "shape" function was used to determine the number of rows and columns in the dataset.

5.2 Dataset Information

```
df.info()
```

The "info()" function was used to examine the number of entries, column names, non-null values, and data types.

The analysis showed that the dataset contains numerical variables such as pH, Temprature, Taste, Odor, Fat, Turbidity, and Colour, while Grade is a categorical variable.

[INSERT SCREENSHOT OF df.info() OUTPUT HERE]

5.3 Statistical Summary

```
df.describe()
```

The "describe()" function was used to obtain summary statistics such as count, mean, standard deviation, minimum, maximum, and quartile values for numerical variables.

This helped in understanding the central tendency and spread of the numerical milk quality parameters.

6. Data Cleaning

Data cleaning was performed to improve the quality and consistency of the dataset before exploratory analysis.

The main cleaning activities included checking missing values, duplicate records, and data types.

6.1 Checking Missing Values

```
df.isnull().sum()
```

The above code was used to identify the number of missing values in each column.

Missing-value checking is important because missing observations can affect statistical calculations, visualizations, and machine learning models.

[INSERT SCREENSHOT OF MISSING VALUE OUTPUT HERE]

Result

The missing-value analysis was performed for all columns. The results were reviewed before proceeding with further analysis.

6.2 Checking Duplicate Records

```
df.duplicated().sum()
```

Duplicate records were checked because repeated observations can influence statistical analysis and may result in biased conclusions.

If duplicate records were identified, they were removed using:

```
df = df.drop_duplicates()
```

The dataset was then checked again to confirm that duplicate records had been handled appropriately.

6.3 Checking Data Types

```
df.dtypes
```

The data types of all columns were examined to ensure that numerical variables were stored as numerical values and the target variable was appropriately treated as categorical data.

The observed data types included:

- pH – float
- Temprature – integer
- Taste – integer
- Odor – integer
- Fat – integer
- Turbidity – integer
- Colour – integer
- Grade – object

7. Data Preprocessing

After inspection and cleaning, the dataset was prepared for exploratory analysis.

The following preprocessing activities were considered:

- Checking missing values

- Checking duplicate records
- Verifying data types
- Identifying numerical and categorical variables
- Ensuring consistency of the dataset

The cleaned dataset was then used for exploratory data analysis.

8. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the characteristics, distributions, and relationships within the milk quality dataset.

Summary statistics and visualizations were used to identify important patterns and potential relationships among the variables.

9. Visualization 1 – Distribution of Numerical Variables

Histograms were used to examine the distribution of numerical milk quality parameters.

```
import matplotlib.pyplot as plt

df.hist(figsize=(12, 8))
plt.tight_layout()
plt.show()
```

Interpretation

The histograms provide an overview of how the numerical milk quality parameters are distributed. They help identify the general range, concentration, and spread of observations for variables such as pH, temperature, fat, turbidity, and colour.

10. Visualization 2 – Correlation Heatmap

A correlation heatmap was used to examine relationships among numerical variables.

```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10, 6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap of Milk Quality Parameters")
plt.show()
```

Interpretation

The correlation heatmap provides a visual representation of the relationships between numerical milk quality parameters. Positive correlation indicates that two variables tend to increase together, while negative correlation indicates an inverse relationship.

The heatmap helps identify variables that may have stronger relationships and therefore may be useful for further statistical analysis and machine learning.

11. Visualization 3 – Milk Quality Grade Distribution

The distribution of the target variable, Grade, was visualized to understand the number of observations belonging to each quality category.

```
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x="Grade")
plt.title("Distribution of Milk Quality Grades")
plt.xlabel("Milk Quality Grade")
plt.ylabel("Count")
plt.show()
```

Interpretation

The grade distribution provides an overview of how the milk samples are categorized into different quality levels. This information is important for understanding the target variable before applying classification algorithms.

12. Additional Visualization – Box Plot

Box plots can be used to identify the spread of numerical variables and possible outliers.

```
plt.figure(figsize=(12, 6))
sns.boxplot(data=df.select_dtypes(include="number"))
plt.xticks(rotation=45)
plt.title("Box Plot of Numerical Milk Quality Parameters")
plt.show()
```

Interpretation

The box plot helps identify the median, spread, and possible extreme observations in the numerical variables. Such observations can be investigated further to determine whether they represent genuine samples or potential data-quality issues.

13. Key Exploratory Insights

The exploratory analysis provided the following initial insights:

1. The dataset contains multiple measurable characteristics that can be used to study milk quality.
2. The variables have different ranges and distributions, which makes individual variable analysis important.
3. The Grade variable provides the overall quality classification of milk samples.
4. Correlation analysis helps identify relationships between numerical milk quality parameters.
5. Visualization provides a clearer understanding of data distributions than numerical summaries alone.
6. The analysis provides a foundation for conducting statistical hypothesis testing.

7. The cleaned and explored dataset can be further used for machine learning-based milk quality prediction.

14. Rationale Behind Data Cleaning Methods

Each data-cleaning step was performed for a specific reason.

Missing Values

Missing values were checked because incomplete observations can affect calculations and model training. The appropriate treatment depends on the number and nature of missing observations.

Duplicate Records

Duplicate records were checked because repeated observations may give excessive importance to the same data point and can influence analysis results.

Data Types

Data types were verified to ensure that numerical calculations could be performed correctly and categorical variables could be handled appropriately.

Data Consistency

The dataset was inspected for inconsistent or inappropriate values to ensure that the data used for analysis was reliable and suitable for further processing.

15. Tools and Technologies Used

The following tools and Python libraries were used:

Python

Python was used as the primary programming language for data analysis.

Pandas

Pandas was used for data loading, manipulation, inspection, and cleaning.

NumPy

NumPy was used for numerical operations and data processing.

Matplotlib

Matplotlib was used to create data visualizations.

Seaborn

Seaborn was used to create statistical visualizations such as heatmaps, count plots, and box plots.

Google Colab

Google Colab was used as the development environment for executing Python code and analyzing the dataset.

16. Overall Findings

The Data Acquisition, Cleaning, and Exploratory Analysis stages provided an initial understanding of the milk quality dataset.

The dataset was successfully loaded and inspected using Python. Its structure, variables, data types, and statistical characteristics were examined. Missing values and duplicate records were checked, and the data was prepared for further analysis.

The visualizations helped identify distributions, relationships, and patterns among milk quality parameters. The correlation analysis provided additional information about relationships between numerical variables.

These findings provide a strong foundation for the next stages of the project, particularly statistical hypothesis testing and machine learning-based milk quality prediction.

17. Conclusion

The Week 1 task successfully demonstrated the fundamental stages of a Data Science project, including data acquisition, data cleaning, preprocessing, and exploratory data analysis.

The Milk Quality Dataset was loaded and examined using Python and Pandas. Data-quality checks were performed for missing values, duplicate records, and data types. Summary statistics were used to understand the numerical variables, while visualizations were created to explore distributions and relationships.

The analysis provided useful initial insights into the milk quality parameters and prepared the dataset for further statistical and machine learning analysis.

Overall, this task established a strong foundation for the subsequent stages of the Data Science project.

18. References

1. Python Documentation – Python programming language documentation.
2. Pandas Documentation – Data manipulation and analysis using Python.
3. NumPy Documentation – Numerical computing with Python.
4. Matplotlib Documentation – Data visualization using Python.
5. Seaborn Documentation – Statistical data visualization.
6. Scikit-learn Documentation – Machine learning and model evaluation.
7. Milk Quality Dataset – Publicly available dataset used for the project.

In [1]:
```python
import pandas as pd
df = pd.read_csv("milknew.csv")
print(df.head())
```

```
    pH  Temprature  Taste  Odor  Fat   Turbidity  Colour   Grade
0  6.6          35      1     0     1          0     254    high
1  6.6          36      0     1     0          1     253    high
2  8.5          70      1     1     1          1     246     low
3  9.5          34      1     1     0          1     255     low
4  6.6          37      0     0     0          0     255  medium
```

In [2]:
```python
print(df.shape)
print(df.info())
print(df.describe())
```

```
(1059, 8)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1059 entries, 0 to 1058
Data columns (total 8 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   pH          1059 non-null   float64
 1   Temprature  1059 non-null   int64
 2   Taste       1059 non-null   int64
 3   Odor        1059 non-null   int64
 4   Fat         1059 non-null   int64
 5   Turbidity   1059 non-null   int64
 6   Colour      1059 non-null   int64
 7   Grade       1059 non-null   object
dtypes: float64(1), int64(6), object(1)
memory usage: 66.3+ KB
None
                pH   Temprature        Taste         Odor          Fat  \
count  1059.000000  1059.000000  1059.000000  1059.000000  1059.000000
mean      6.630123    44.226629     0.546742     0.432483     0.671388
std       1.399679    10.098364     0.498046     0.495655     0.469930
min       3.000000    34.000000     0.000000     0.000000     0.000000
25%       6.500000    38.000000     0.000000     0.000000     0.000000
50%       6.700000    41.000000     1.000000     0.000000     1.000000
75%       6.800000    45.000000     1.000000     1.000000     1.000000
max       9.500000    90.000000     1.000000     1.000000     1.000000

         Turbidity       Colour
count  1059.000000  1059.000000
mean      0.491029   251.840415
std       0.500156     4.307424
min       0.000000   240.000000
25%       0.000000   250.000000
50%       0.000000   255.000000
75%       1.000000   255.000000
max       1.000000   255.000000
```

In [3]:
```python
# Check missing values
print(df.isnull().sum())
```

```
pH             0
Temprature     0
Taste          0
Odor           0
Fat            0
Turbidity      0
Colour         0
Grade          0
dtype: int64
```

In [4]:
```
# Missing value percentage
print((df.isnull().sum() / len(df)) * 100)
```

```
pH             0.0
Temprature     0.0
Taste          0.0
Odor           0.0
Fat            0.0
Turbidity      0.0
Colour         0.0
Grade          0.0
dtype: float64
```

In [5]:
```
# Check duplicate rows
print("Duplicate rows:", df.duplicated().sum())
```

```
Duplicate rows: 976
```

In [6]:
```
# Remove duplicates
df = df.drop_duplicates()
print("Shape after removing duplicates:", df.shape)
```

```
Shape after removing duplicates: (83, 8)
```

In [7]:
```
# Check data types
print(df.dtypes)
```

```
pH             float64
Temprature       int64
Taste            int64
Odor             int64
Fat              int64
Turbidity        int64
Colour           int64
Grade           object
dtype: object
```

In [8]:
```
print(df.describe())
```

```
              pH  Temprature      Taste       Odor        Fat   Turbidity  \
count  83.000000   83.000000  83.000000  83.000000  83.000000  83.000000
mean    6.668675   43.698795   0.493976   0.397590   0.602410   0.433735
std     0.986856    9.379187   0.503003   0.492375   0.492375   0.498602
min     3.000000   34.000000   0.000000   0.000000   0.000000   0.000000
25%     6.500000   38.000000   0.000000   0.000000   0.000000   0.000000
50%     6.600000   41.000000   0.000000   0.000000   1.000000   0.000000
75%     6.800000   45.000000   1.000000   1.000000   1.000000   1.000000
max     9.500000   90.000000   1.000000   1.000000   1.000000   1.000000

           Colour
count   83.000000
mean   251.313253
std      4.577058
min    240.000000
25%    247.000000
50%    255.000000
75%    255.000000
max    255.000000
```

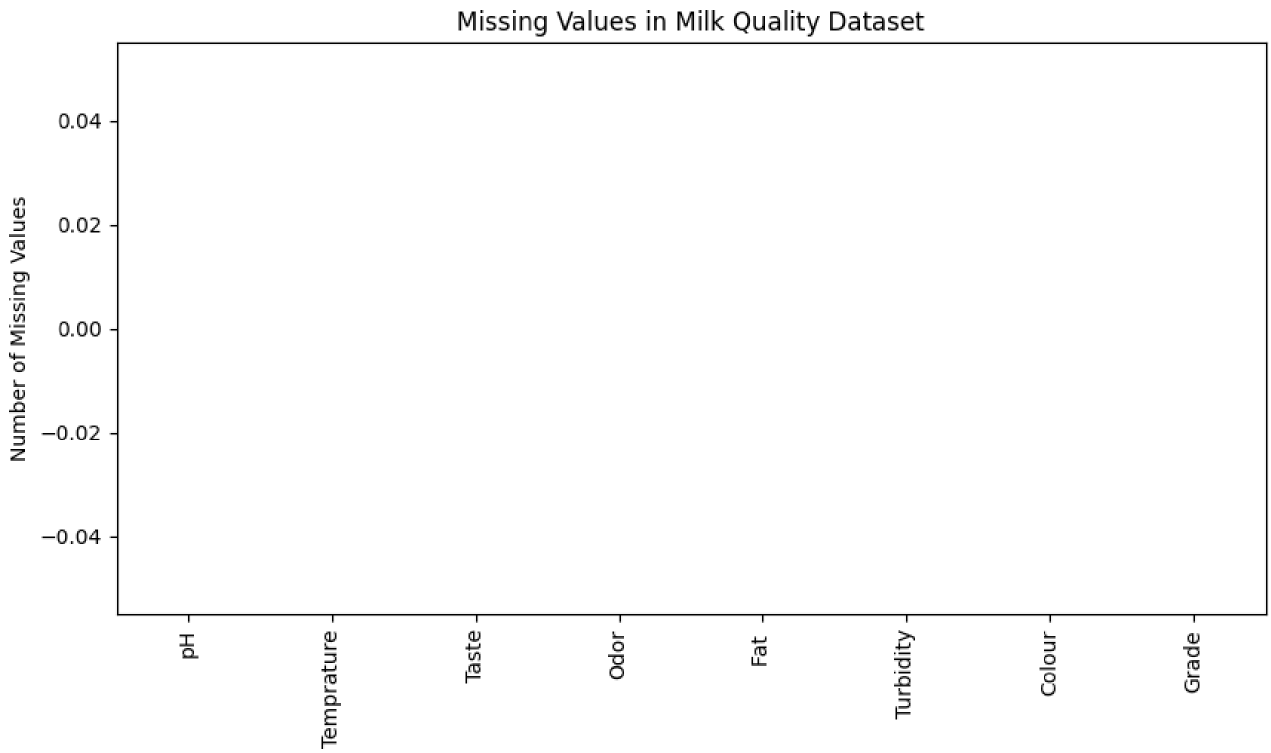
In [9]:

```
import matplotlib.pyplot as plt

missing = df.isnull().sum()

plt.figure(figsize=(10,5))
missing.plot(kind='bar')

plt.title("Missing Values in Milk Quality Dataset")
plt.xlabel("Columns")
plt.ylabel("Number of Missing Values")
plt.show()
```



In [10…

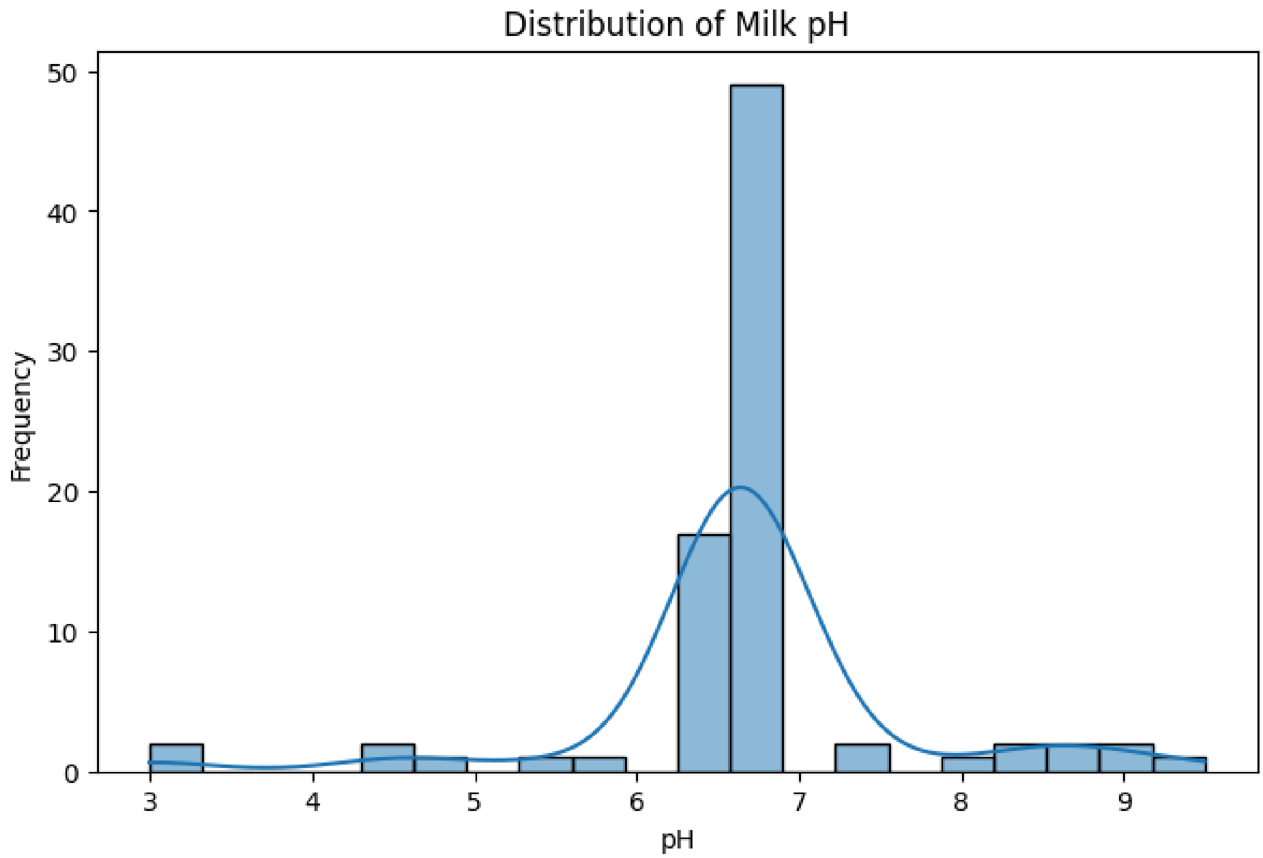
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8,5))

sns.histplot(df['pH'], bins=20, kde=True)

plt.title("Distribution of Milk pH")
plt.xlabel("pH")
plt.ylabel("Frequency")

plt.show()
```



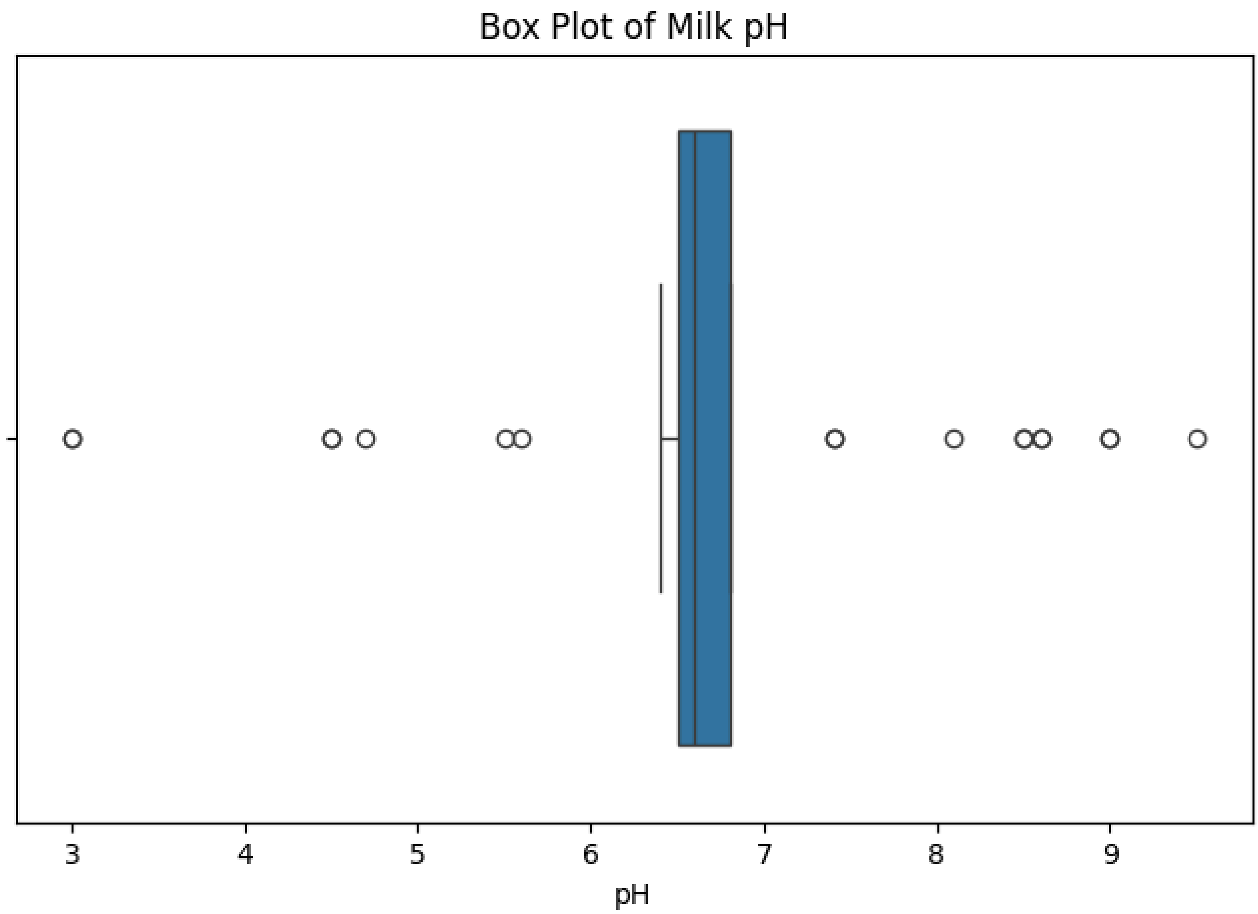
In [11…

```
plt.figure(figsize=(8,5))

sns.boxplot(x=df['pH'])

plt.title("Box Plot of Milk pH")
plt.xlabel("pH")

plt.show()
```



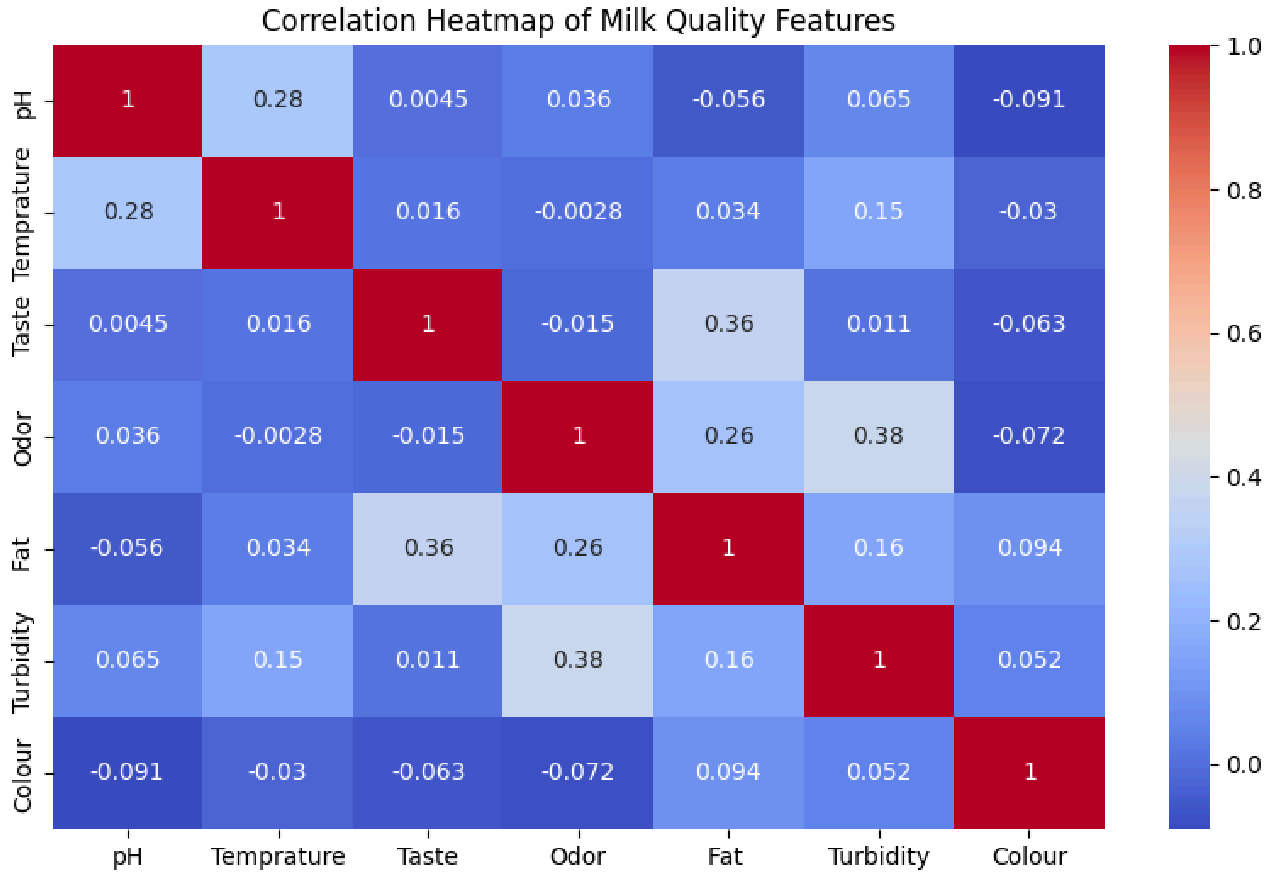
In [12…

```
plt.figure(figsize=(10,6))

correlation = df.select_dtypes(include='number').corr()

sns.heatmap(correlation, annot=True, cmap='coolwarm')

plt.title("Correlation Heatmap of Milk Quality Features")
plt.show()
```



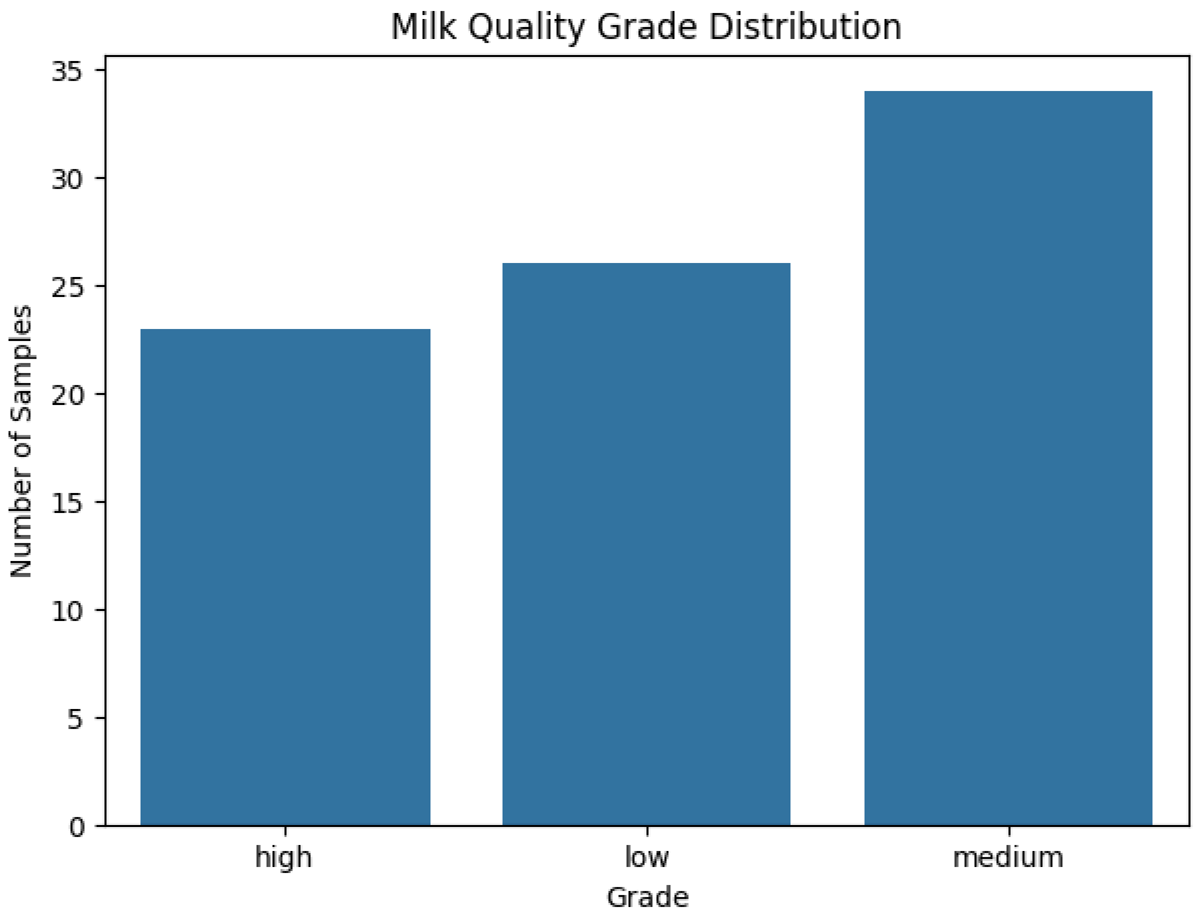
In [13…

```
plt.figure(figsize=(7,5))

sns.countplot(data=df, x='Grade')

plt.title("Milk Quality Grade Distribution")
plt.xlabel("Grade")
plt.ylabel("Number of Samples")

plt.show()
```



In [14…

```
print(df['Grade'].value_counts())
```

```
Grade
medium    34
low       26
high      23
Name: count, dtype: int64
```

In [17…

```
print(df[['pH', 'Temprature', 'Turbidity', 'Colour']].mean())
```

```
pH              6.668675
Temprature     43.698795
Turbidity       0.433735
Colour        251.313253
dtype: float64
```